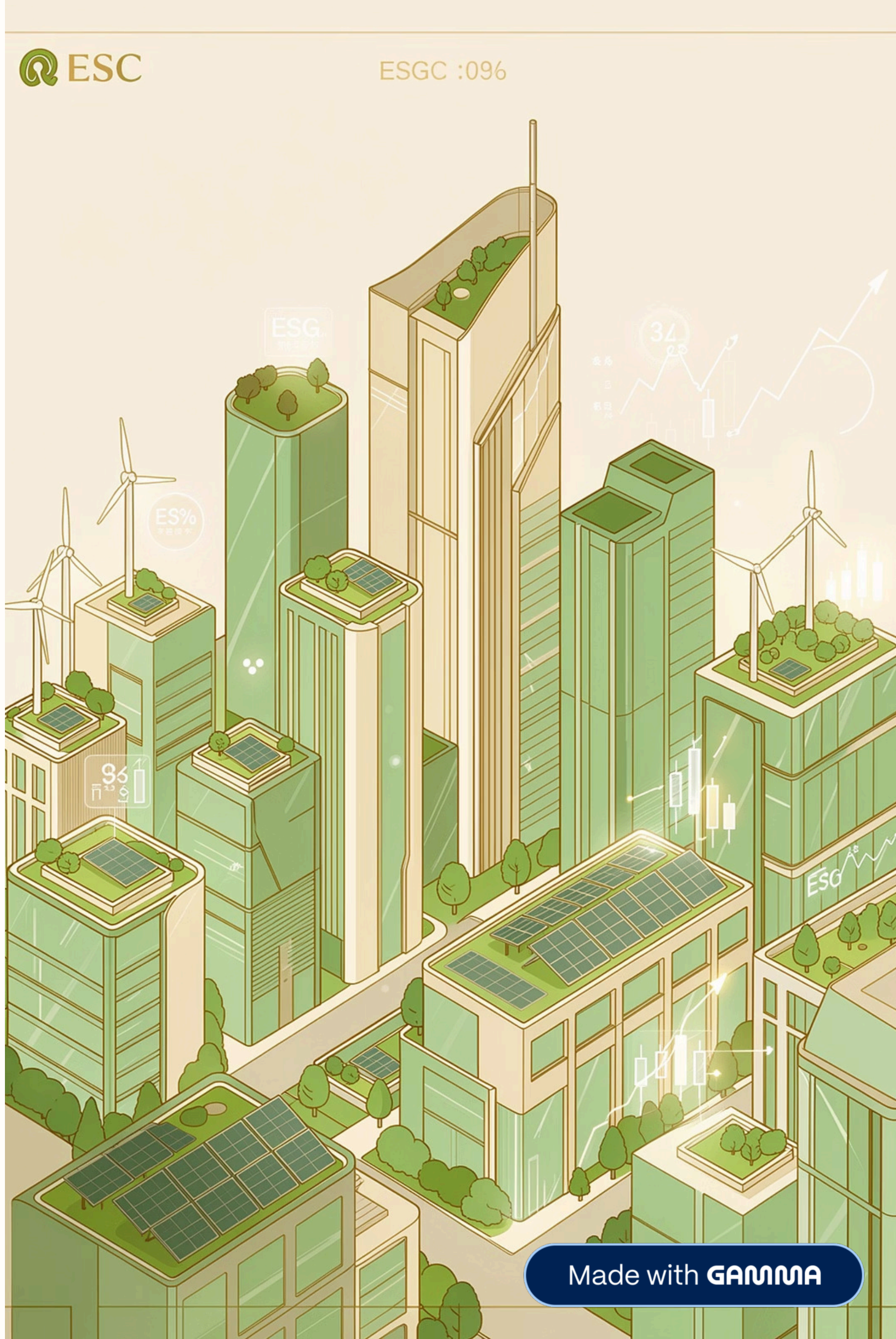
ESG Risk & Investment Analysis

Identifying sustainable, low-risk companies for long-term investment

DATA SCIENCE PROJECT



Project Overview

Objective

Develop an ESG evaluation framework to identify sustainable, low-risk companies for long-term investment.

Problem Statement

Which companies are sustainable and low-risk for long-term investment?

Tools Used

Python · Power BI · Jupyter Notebook

Dataset at a Glance

ESG Scores Dataset — **Kaggle** · 10,000 companies across industries

ESG Score

Overall rating (0–100)

Environmental

Environmental performance

Social

Social responsibility

Governance

Governance quality

Profit Margin

Profitability indicator

Exploratory Data Analysis

Data Cleaning

Removed missing values & duplicates · Standardized ESG format

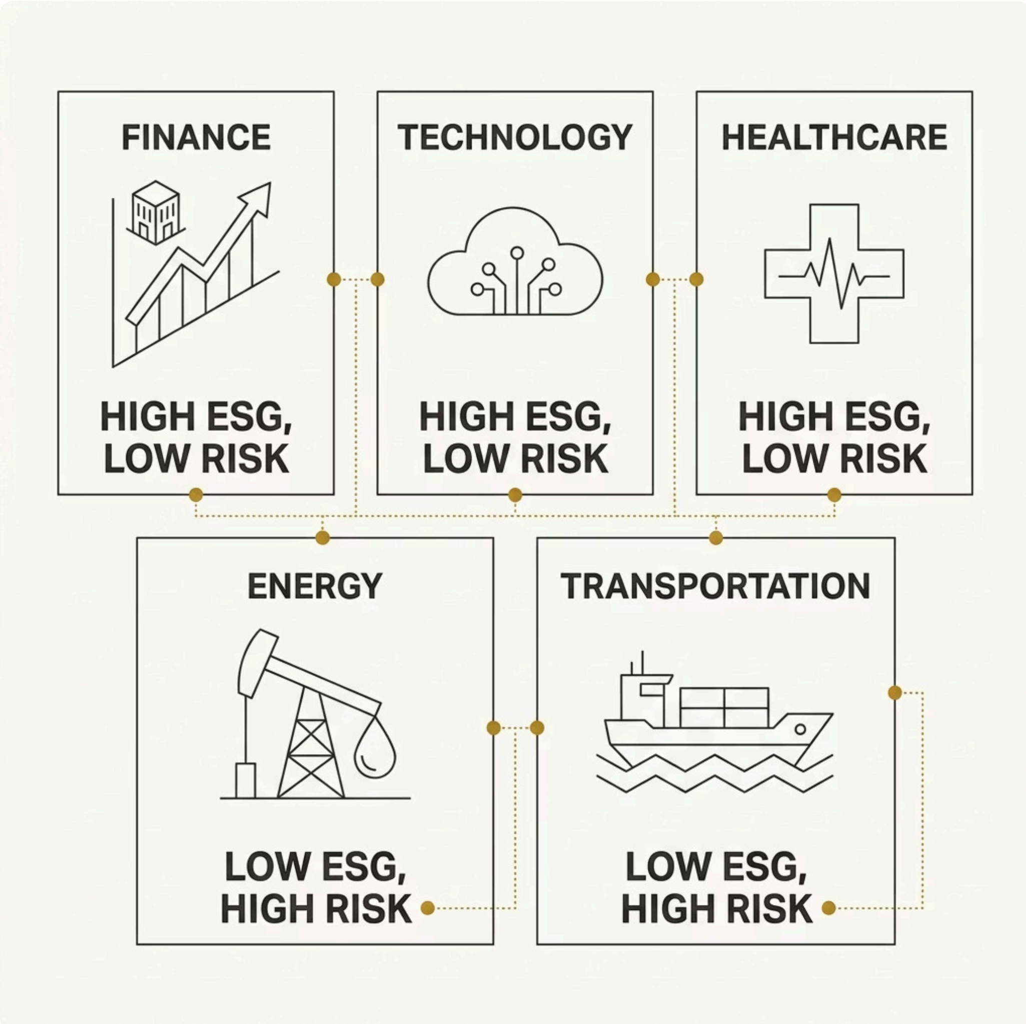
ESG Distribution

Most companies fall in **medium ESG range** — moderate sustainability

ESG vs Profitability

Positive correlation — higher ESG scores → higher profits

Industry ESG Performance



Top Performers

Finance · Technology · Healthcare

Low Performers

Energy · Transportation

> Energy & Transportation show lower ESG scores and higher investment risk

Risk Classification

Companies categorized by ESG score into three risk tiers

🟢 Low Risk	🟡 Medium Risk	🔴 High Risk
ESG ≥ 70	40 ≤ ESG < 70	ESG < 40

Dashboard Insights

55.13

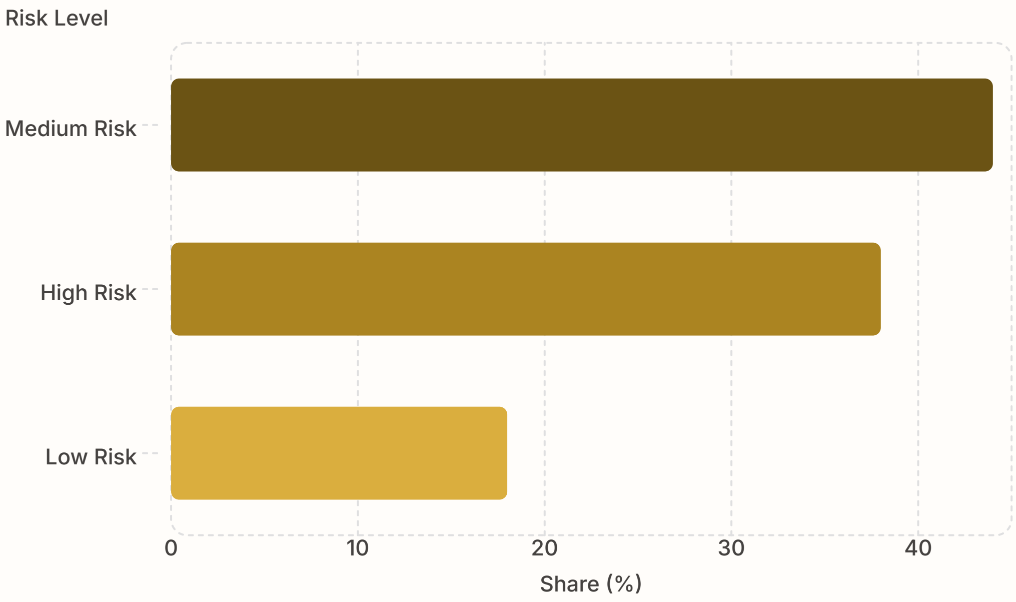
Avg ESG Score

10K

Total Companies

38%

High-Risk Firms



Key Finding

Only **18%** of companies qualify as low-risk investment options — limited safe choices in the market.

Investment Strategy

1 Prioritize Low Risk

Target ESG ≥ 70 · Focus: Finance, Technology, Healthcare

2 Portfolio Allocation

60% Low Risk · 30% Medium Risk · 10% High Risk

3 Risk Management

Monitor ESG scores · Exit companies below ESG 40

Industry-Level Strategy

Energy & Transportation

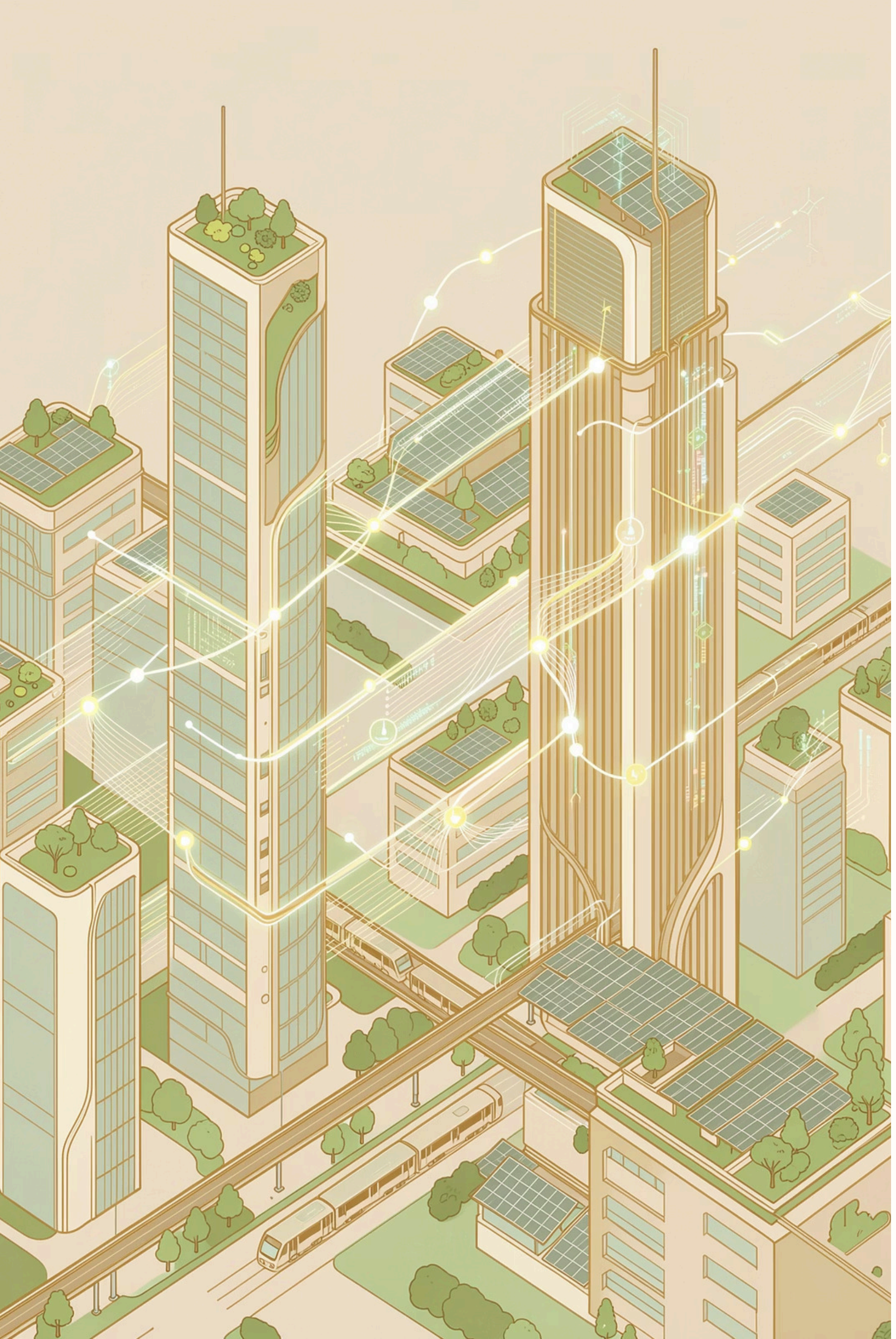
Invest in sustainable practices · Focus on emission reduction & clean technologies

Finance & Technology

Maintain ESG leadership · Use sustainability as a competitive advantage

Profitability Insight

ESG is a **growth driver**, not a cost — strong ESG = better financial outcomes



Conclusion & Future Scope

Key Takeaway

ESG = financial stability & long-term investment potential

Predictive Models

ML-based ESG scoring & risk prediction

Time-Series Analysis

ESG trends + stock price integration